

教學大綱 Syllabus

科目名稱：投資與資產組合

Course Name:
Investment & Portfolio Management

📅 學年學期：115-1
Fall Semester, 2026

📄 科目代碼：301813001
Course No.301813001



修別：
選

Type of
Credit:
Elective

3.0

學分數
Credit(s)

30

預收人
數

Number
of
Students

課程資料

Course Details

開課單位：國貿三甲、國貿三乙、國貿四甲、
國貿四乙

Course Department: International Business/B/3&4

授課老師：鍾令德

Instructor: CHUNG LING TAK

先修科目：無

Prerequisite(N/A)

上課時間：五234

Session: fri09-12

課程簡介

Course Description

This **advanced** undergraduate course covers the fundamental principles of investment, with an emphasis on theory and application. The first part of the course will give an overview of the financial market and discuss optimal portfolio choice under the modern portfolio theory. We will then discuss the equilibrium implications of optimal portfolio choice and derive the capital asset pricing model (CAPM). Next, we will look into the arbitrage pricing theory (APT) and factor investing. The last part will cover other asset classes, performance evaluations, and risk management.

We will extensively use Python to analyze financial data and backtest portfolio strategies. There are no prerequisites for the course, but students are expected to have a strong background in statistics and calculus.

This course is **more difficult and demanding** than the **Data-driven Investment Analysis** course offered in the same semester.

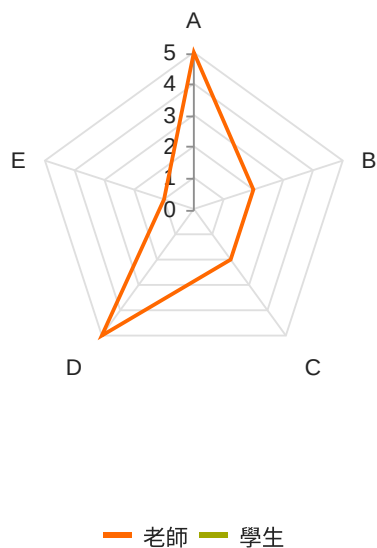
This is a **distance-learning (遠距課程)** course. We will only meet in the classroom for Week 1 (Course Overview).

核心能力分析圖



Core Competence Analysis Chart

雷達圖



能力項目說明

- A. 專業知識之理論基礎 (Theoretical foundations of professional knowledge)
- B. 理解與辨別道德、社會責任與永續議題To understand and identify ethical, socially responsible, and sustainability issues
- C. 溝通能力Communication skills
- D. 問題解決能力Problem-solving skills
- E. 團隊合作與領導能力Teamwork and leadership skills

課程目標與學習成效

Course Objectives & Learning Outcomes

- 1. To understand the fundamental theories of portfolio choice and asset pricing.
- 2. To understand and apply portfolio management techniques.
- 3. To understand the investment framework and conduct performance evaluation and risk management.
- 4. To acquire practical experience in working with financial data.
- 5. To develop skills in reading academic and practitioner journals.
- 6. To gain insights about trends in investment and portfolio management.



級分說明 Score Description:

1關聯性最弱，2關聯性稍弱，3關聯性中等，4關聯性稍強，5關聯性最強。(1-5非百分比，而是Rank)
 1 indicates weakest relevance, 2 indicates slightly weaker relevance, 3 indicates moderate relevance, 4 indicates slightly stronger relevance, and 5 indicates strongest relevance. (1-5 does not represent a certain percentage but indicates the rank).

課程屬性項目 Course Characteristics Items	級分 (1~5) Ranking	說明 Description
人文關懷課程 Humanity and Care Course	1	課程教學內容涉及學生人文關懷素養與態度之培養。 The course contents include the development of students' abilities and appreciation for humanistic care.
跨領域課程 Interdisciplinary Course	5	課程教學包含不同學門、領域之內容。 The courses include contents from different disciplines and fields.
社會參與課程 Social Engagement Course	1	課程教學內容涉及社會參與、服務等。 The course contents include social engagement and services.
總整課程 Capstone Course	5	係系統整與深化學生歷年所學，並運用於實務操作與討論，藉以檢視和評估學生的整體學習成效。 The courses are designed to integrate and intensify students' academic results from previous years and apply them in practice and discussions for the purpose of reviewing and evaluating the overall effectiveness of students' learning.
永續課程 SDGs	2	課程主軸明確探討永續性之統整概念及/或聯合國永續發展目標及/或「生態」與「社會/文化/經濟」系統之間的相互依賴性。關聯度級分請點選連結說明。 A unit of instruction that has a primary and explicit focus on the integrated concept of sustainability, and/or UN Sustainable Development Goals, and/or the interdependence of ecological and social/cultural/economic systems. Please refer to the link of relevance score.
資訊判讀課程 Information Literacy Course	5	課程內容旨在教授媒體、網路相關專業資訊之真偽辨別與判讀能力。 The course would provide skills to evaluate the quality of information found on the media and the Internet.
AI 素養課程 AI Competence Course	2	將 AI 知能運用融入課程，引導師生與 AI 共學、協作，多元培養學生 AI 素養。 Integrate the application of AI knowledge into the curriculum, lead teachers and students to learn and collaborate with AI, and cultivate students' AI literacy.



每周課程進度與作業要求

Course Schedule & Requirements

週次 Week	課程主題 Topic	課程內容與指定閱讀 Content and Reading Assignment	教學活動與作業 Teaching Activities and Homework	學習投入時間 Student workload expectation	
				課堂講授 In-class Hours	課程前後 Outside-of-class Hours
1	Course overview and quantitative tools (11-Sep-2026, classroom)	Lecture notes	Class 1	3	6
2	Financial market basics (18-Sep-2026, pre-recorded)	BKM Chapters 1, 2, 3	Class 2	3	6
3	Index replications and passive investments (25-Sep-2026, pre-recorded)	BKM Chapter 5	Class 3	3	6
4	Portfolio choices I (2-Oct-2026, pre-recorded)	BKM Chapter 6	Class 4	3	6
5	Portfolio choices II (9-Oct-2026, pre-recorded)	BKM Chapter 7	Class 5	3	6
6	Applications of portfolio choices (16-Oct-2026, pre-recorded)	BKM Chapters 7, 8	Class 6	3	6
7	Midterm examination (23-Oct-2026, online)				6
8	CAPM (30-Oct-2026, pre-recorded)	BKM Chapter 9	Class 7	3	6



9	Applications of CAPM (6-Nov-2026, pre-recorded)	BKM Chapters 11, 13 and journal articles	Class 8	3	6
10	Multifactor models (13-Nov-2026, pre-recorded)	BKM Chapter 10	Class 9	3	6
11	Performance evaluation and risk management (20-Nov-2026, pre-recorded)	BKM Chapters 20, 22, 23, 24	Class 10	3	6
12	Factor investing I (27-Nov-2026, pre-recorded)	Lecture notes and journal articles	Class 11	3	6
13	Factor investing II (3-Dec-2026, pre-recorded)	Lecture notes and journal articles	Class 12	3	6
14	Fund industry (11-Dec-2026, pre-recorded)	BKM Chapter 4	Class 13	3	6
15	Final examination (18-Dec-2026, online)		Final examination		6

授課方式

Teaching Approach

60%

講述
Lecture

0%

討論
Discussion

0%

小組
活動
Group
activity

40%

數位
學習
E-
learning

0%

其他：
Others:

評量工具與策略、評分標準成效

Evaluation Criteria

Assessments:

1. Midterm examination (40%).
2. Final examination (40%).
3. Participation on the Moodle's course forum (20%).



指定/參考書目

Textbook & References

Main textbook:

Investments, 11th Edition by Zvi Bodie, Alex Kane, and Alan Marcus, McGraw-Hill, 2019. (BKM)

Additional references:

Modern Portfolio Theory and Investment Analysis, 9th Edition by Edwin Elton, Martin Gruber, Stephen Brown, and William Goetzmann, Wiley, 2014.

Asset Management, A Systematic Approach to Factor Investing, 1st Edition by Andrew Ang, Oxford University Press, 2014.

已申請之圖書館指定參考書目

書名 Book Title	作者 Author	出版年 Publish Year	出版者 Publisher	ISBN	館藏來源*	備註 Note
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課程AI能力關聯

Course-AI Competency Relevance

級分說明 Score Description:

- 0：無關聯，課程內容、活動及評量均未涉及。
- 1：極低關聯，僅於課堂中簡單提及，未安排學習活動。
- 2：低度關聯，少數單元或活動涉及，但非主要教學內容。
- 3：中度關聯，明確納入部分教學活動、作業或評量。
- 4：高度關聯，為重要學習內容，融入多數教學活動或評量。
- 5：核心關聯，為課程核心學習目標，且有系統性教學與評量。

- 0 : No relevance: The course content, learning activities, and assessments do not address this competency.
- 1 : Very low relevance: The competency is only briefly mentioned in class, with no related learning activities.
- 2 : Low relevance: The competency is addressed in a small number of units or activities but is not a major component of the course.
- 3 : Moderate relevance: The competency is explicitly incorporated into some learning activities, assignments, or assessments.
- 4 : High relevance: The competency is an important learning component and is integrated into most learning activities or assessments.
- 5 : Core relevance: The competency is a core learning objective of the course and is addressed through systematic instruction and assessment.



課程AI能力項目 Course-AI Competency Items	級分 (0~5) Ranking	說明 Description
能力A：教授學生運用 AI 學習的能力 Competency A: Teaching Students to Use AI for Learning	2	本課程培養學生依專業或學習需求，理解、選用及運用AI工具，進行資訊整理、資料產出、提問與自主學習的程度。 This course develops students' ability to understand, select, and use AI tools according to their disciplinary or learning needs for information organization, data or content generation, inquiry, and self-directed learning.
能力B：教授學生負責 任使用 AI 的能力 Competency B: Teaching Students to Use AI Responsibly	2	本課程培養學生遵循學術誠信、智慧財產權、個人資料及相關倫理規範，並辨識AI偏誤、錯誤資訊與資安風險的程度。 This course develops students' ability to comply with academic integrity, intellectual property, personal data protection, and relevant ethical requirements, as well as to identify AI bias, misinformation, and cybersecurity risks.
能力C：教授學生運用 AI解決問題的能力 Competency C: Teaching Students to Use AI for Problem-Solving	2	本課程培養學生運用AI分析問題、整合專業知識、優化流程，並透過人機協作提出、驗證及修正解決方案的程度。 This course develops students' ability to use AI to analyze problems, integrate disciplinary knowledge, optimize processes, and propose, validate, and refine solutions through human-AI collaboration.
能力D：教授學生AI無 法取代的能力 Competency D: Developing Human- Centered Competencies That AI Cannot Replace	3	本課程培養學生自主判斷、承擔責任、溝通協作、同理關懷、多元思考及創意實踐等以人為本能力的程度。 This course develops students' human-centered competencies, including independent judgment, accountability, communication and collaboration, empathy and care, diverse perspectives, and creative practice.

本課程可否使用 生成式AI工具

Course Policies on the Use of

Generative AI Tools

完全開放使用

Completely Permitted to Use

課程相關連結

Course Related Links





課程附件

Course Attachments

課程進行中，使用智慧型手機、平板等隨身設備

To Use Smart Devices During the Class

是
Yes

授課教師Office Hours及地點

Office Hours & Office Location

TBA

教學助理基本資料

Teaching Assistant Information

TBA

